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# Asymmetric impact of economic policy uncertainty on cryptocurrency market: Evidence from NARDL approach

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## Abstract

The study examines the relationship between cryptocurrency market and economic policy uncertainty (EPU) by concentrating on the top five cryptocurrencies assessed by market capitalization. We investigate long and short-run effects of global EPU on the returns of Bitcoin, Ethereum, Tether, Binance coin and Ripple, along with ascertaining asymmetries, through the NARDL (Non-Linear Autoregressive Distributed Lag) model, for the period ranging from September 2017 to August 2022. In the long run, except for Tether, all other cryptocurrencies are negatively affected by EPU, challenging the safe hedge properties of cryptocurrencies. However, Bitcoin, Ethereum, Binance coin and Ripple display positive relations with growing EPU in short run, reflecting their safe haven attributes over a shorter horizon. Additionally, these cryptocurrencies exhibit asymmetries in the short run. Further, we analyse the effect of US, Chinese and Indian EPU on bitcoin returns to understand the effect of EPU shocks of advanced and emerging economies. US EPU has long-run negative effects on bitcoin market, whereas Indian and Chinese EPU have no significant impact in long term. Moreover, our analysis also establishes the detrimental impact of the Indian government's crypto tax policy on bitcoin. The study has relevance for current and potential

## Keywords

Asymmetry; Bitcoin; Cryptocurrency; EPU; NARDL; Safe haven

## JEL classification

C22; E44; G15; G18

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## Introduction

The cryptocurrency market has grown astonishingly over the past decade and has piqued the interest of investors, governments, financial institutions and academicians worldwide, ever since the inception of “Bitcoin” in 2009 (Colon et al., 2021). Ahead of the 2008 global crisis, cryptocurrency market, particularly bitcoin, began to develop. Due to the global crisis, individuals experienced tremendous losses raising their scepticism about the financial system (Aharon et al., 2022; Demir et al., 2018). The lack of certainty sparked a need for a decentralized currency independent of government policy and banking regulations, fuelling the development of the cryptocurrency market. Thus, in order to combat the fear and ambiguity in the economic system, Bitcoin was created (Wang et al., 2019).

According to the information from the coingecko website, the cryptocurrency market has dramatically expanded from a 1-million-dollar market capitalization in 2013 to crossing 3 trillion dollars in November 2021. The dominance of bitcoin has dropped immensely from almost 90% in October 2014 to 38% in October 2022, approached even less than half; this is attributable to the emergence of altcoins (cryptocurrencies other than bitcoin) in the cryptocurrency industry. Later in 2014, Tether was introduced as the first stablecoin; these are the digital currencies whose value is intended to be tied to a benchmark asset, which is US dollar in the case of Tether (Papadamou et al., 2021). Launched in 2015, Ethereum is presently the second dominant cryptocurrency after bitcoin, according to market capitalization. As of November 2022, more than 20,000 cryptocurrencies are available in the market,

**Economic Policy Uncertainty** (EPU) has been observed as one of the determinants affecting returns and volatilities of cryptocurrencies, particularly Bitcoin (Bouri et al., 2021; Mokni, 2021; Panagiotidis et al., 2018, 2019; Qin et al., 2021). It is well known that the economic development of a country is heavily influenced by its economic policies, so any ambiguities in these may inhibit the economy's growth. The heightened EPU raises investors' risk-aversion attitude, consequently impeding investment flows. Therefore, investors are pulled to devise risk mitigation strategies during times of financial crisis or unrest (Haq et al., 2021). As a severe economic downturn was sparked by the intensification of apprehensions and anxieties about policy uncertainty during 2008–09, EPU has been linked to these distress situations (Baker et al., 2016), and during that period only, Nakamoto created the digital currency “Bitcoin”.

Over time, cryptocurrencies have emerged more like as an asset class than a currency (Bianchi, 2020; Ram, 2019). Due to the enormous market capitalization and transactional dynamics, they have developed as unique financial assets accepted worldwide without restrictions. Though grounded on the new technology of blockchain, which is not yet fully grasped, cryptocurrency still functions as equivalent to other conventional assets in its existing form (Liu & Tsyvinski, 2021). It has been demonstrated in earlier studies that EPU considerably impacts asset returns, owing to the “Wait & See” effect (Brogaard & Detzel, 2015; Liang et al., 2020; Pástor & Veronesi, 2013). Thus, EPU as an influencing factor deserves greater attention as a possible factor affecting the returns of cryptocurrencies.

For studying EPU's impact on cryptocurrencies, Yen and Cheng (2021) develop two arguments related to “Fear” and “Hedging”. The first states that investors withdraw money from the cryptocurrency market due to the fear and anxiety generated by EPU, bringing a downturn in the crypto market. In contrast, another case relates to the increased investment in cryptocurrency, considering it a safe haven during uncertain times. On a similar note, Qin et al. (2021) state that global EPU (GEPU) influences bitcoin prices through five mechanisms related to its demand and supply. Firstly, it appears as a hedging mechanism to mitigate risks in highly uncertain periods; second, it indirectly influences bitcoin prices by impacting other assets.

as high EPU impedes technological advancements and lastly, national policies regarding the digital currency market also impact the bitcoin market. Thus, it is evident that EPU has some bearing on the cryptocurrency market.

Academic articles about the influences of uncertainty indexes on cryptocurrency market are available in literature. However, most of the previous literature has largely focussed on a single cryptocurrency- Bitcoin, that captures the largest crypto market share; few research articles study other cryptocurrencies (altcoins). Our study attempts to expand this under-researched area by considering the association between EPU and the leading five cryptocurrencies (Bitcoin, Ethereum, Tether, Binance coin and Ripple) as per the market capitalization. Also, so far, no study has looked at the asymmetric impact of EPU on the leading cryptocurrencies utilizing NARDL. This paper investigates the long and short-run cointegrating relationship between EPU and the cryptocurrency market and explores asymmetric effects. The studies assessing the effect of country-specific EPU shocks on cryptocurrencies are limited (Cheng & Yen, 2020; Yen & Cheng, 2021). Moreover, no study has looked at the long and short-run impact of Indian EPU on cryptocurrency market.

Our research shows that, over the long term, rising global EPU adversely impacts the returns of Bitcoin, Ethereum, Binance Coin, and Ripple. In contrast, the declining EPU has a favourable effect, showing that investors regain confidence in the cryptocurrency market after uncertainty subsides. Thus, it can be argued that these assets do not behave as safe-haven assets in the long term. Tether is positively impacted by rising EPU due to its stablecoin status and acting like safe haven asset in long run. Bitcoin, Ethereum, Binance coin and Ripple are positively impacted in the short run, indicating safe haven property throughout the shorter horizon. Furthermore, asymmetries are found significant only in the short run and not in long run. The results of the country-specific EPU-bitcoin nexus provide that US EPU has a long-run negative impact on bitcoin returns, as US has the dominant role in the global economy. In contrast, India and China EPU have no significant long-run effect on Bitcoin. The decentralized and global nature of the cryptocurrency market could help explain the insignificant long-run effect of Chinese and Indian EPU indices on cryptocurrency return. Yet, it is susceptible to economic policy

Article preview

Abstract

**Introduction**

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also shows that the Indian government's recent strategy of taxing cryptocurrencies has had a negative impact on Bitcoin market and led to a decline in the prices of cryptocurrencies.

This article is continued as follows. Section 2 delivers an overview of related studies. Section 3 provides the information about data. The methodology has been detailed in section 4. Section 5 reports the empirical findings and discussion. Section 6 includes conclusion.

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## Section snippets

### Literature review

In recent years, numerous empirical studies in finance literature have been devoted to a detailed investigation of cryptocurrencies. While analyzing the evolutionary trends of cryptocurrency studies, Jiang, Li, et al. (2021) state that the present research on cryptocurrencies can be divided into two groups. The first one consists of technological advancements that underpin crypto and another category is of its economic and financial implications. Our study belongs to the latter category. …

### Data and descriptive statistics

The study's purpose is to check cointegrating relationship between the cryptocurrency market and EPU. We have considered the

examined in our study are Bitcoin (BTC), Ethereum (ETH), Tether (USDT), Binance coin (BNB) and Ripple (XRP). The combined market share of these five cryptocurrencies is greater than …

## Methodology

The present study applies Non-Linear Autoregressive Distributed Lag (NARDL) model, devised by Shin et al. (2014), for examining the asymmetric influence of EPU on cryptocurrency markets and exploring cointegrating relationships. This approach builds on the long-established Linear ARDL method proposed by Pesaran, M. H., & Shin (1999). By creating a single equation model, NARDL delivers all the benefits of conventional ARDL and aids in capturing the long- and short-run asymmetric relationship …

## Tests for stationarity

The NARDL model states that it is only accurate and useful when variables are a mixture of different integration orders I (1) or I (0) and not I (2) (Shin et al., 2014). Therefore, it is required to test for stationarity to ascertain whether all variables are qualified to be contained in our model before proceeding for the empirical estimation. ADF and PP tests are applied to detect the presence of unit roots. Variables are examined with various constant and trend specifications. Results of …

## Conclusion

Our research adds to the growing body of knowledge about digital currencies and economic uncertainties in academia. In this paper, the risk-hedging property of cryptocurrencies is examined under the circumstances of global uncertainty being bred out of policymaking. We apply NARDL approach to assess the asymmetric impact of EPU on returns of leading cryptocurrencies - Bitcoin, Ethereum, Tether, Binance coin and Ripple, besides detecting long and short-term relationships. Over a long period, …

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## Declaration of competing interest

The authors have no conflicts of interest to declare.

All authors have seen and agree with the contents of the manuscript, and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication. …

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